

SAGNIK CHANDRA

MSc in Data Science and AI, RKMVERI, Belur

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SUMMARY

Strong foundation in mathematics, machine learning, and deep learning with hands-on experience in NLP, large language models, and distributed data processing. Interested in bridging theoretical concepts with practical AI systems and building scalable, data-driven solutions for real-world applications.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute

MSc in Data Science and Artificial Intelligence, 2025–27

University of Calcutta

BSc in Mathematics, 2020–23
CGPA – 7.468

EXPERIENCE

Research Intern – Indian Statistical Institute

May 2026 – Ongoing

- Analyzed the existence of the "Lost in the Middle" phenomenon in LLMs on factoid texts as defined in the source publication, on custom datasets defined from NaturalQuestions using Llama 3.1 8b instruct.
- Currently working on defining an extended experimental validation of the Lost in the Middle phenomenon on complete RAG pipelines on non-factoid texts using a modified MS MARCO 2.1 dataset from the TREC RAG 2024 benchmark.

Data Science Intern – DeepThought CultureTech Ventures

Oct 2024 – Jul 2025

- Led and contributed to 10+ cross-functional initiatives spanning AI automation, CRM optimization, data operations, recruitment, UX research, and business growth.
- Built AI-powered workflows and redesigned CRM/KPI reporting systems, reducing manual effort by 1–4 hours daily, cutting turnaround time from 3 days to 1 day, and improving process efficiency by 60% for 30+ stakeholders.

SKILLS

- Languages:** Python, C, Cypher, R
- Frameworks:** Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, PySpark
- Operating Systems:** Windows, Linux (Ubuntu)
- Tools:** Neo4j, Git, GitHub, Jupyter Notebook

CERTIFICATIONS

- Optimisation Using Genetic Algorithms and Machine Learning using Neural Networks** (IIT Kharagpur, 2026)
- Neural Networks and Deep Learning** (Coursera, 2026)

PROJECTS

Neural Text Style Transfer with Adversarial Learning

RKMVERI

Jan 2026 – May 2026

Built a semi-automated pipeline to rewrite Bengali sentences in the style of five renowned authors without parallel corpora. Trained a BiGRU encoder with a Gradient Reversal Layer (GRL) and a style-conditioned GRU decoder to disentangle author style and generate target-style text, reducing discriminator accuracy from ~65% to near chance (~20%). Curated an 18,779-sentence corpus spanning five authors (Tagore, Sarat Chandra, Satyajit Ray, Bibhutibhushan, Sunil Gangopadhyay) using a pdfplumber extraction pipeline with Tesseract OCR fallback for scanned text. Employed a warm-up and dynamic gradient-reversal schedule during adversarial training, and top-k sampling with temperature scaling and repetition penalty during autoregressive generation to produce fluent, style-transferred output.

AcademicLens – Citation Graph Mining at Scale

RKMVERI

Jan 2026 – Ongoing

Built a large-scale academic intelligence system over 10M+ OpenAlex research papers using PySpark and Neo4j to analyze research influence, collaborations, and topic trends. Designed a distributed ETL pipeline with a multi-hop Author–Paper–Citation–Topic graph and applied PageRank for influence ranking.

Stellar Object Classification using Tree-Based Ensemble Models

RKMVERI

Oct 2025 – Dec 2025

Developed a supervised multi-class classification system to identify galaxies, quasars (QSOs), and stars from the Sloan Digital Sky Survey (SDSS DR18) using tree-based ensemble learning models. Evaluated and compared Random Forest, XGBoost, LightGBM, CatBoost, and AdaBoost under class imbalance and correlated features to optimize classification accuracy and per-class recall. Conducted feature engineering to remove highly correlated ($|r| > 0.9$) and administrative columns, and used Optuna for Bayesian hyperparameter tuning across all models. CatBoost and XGBoost emerged as top performers, achieving test accuracies above 99% and macro F1 scores of 0.987, with QSO identified as the most challenging class due to its photometric similarity to stars.

Traffic-Aware Single-Drone Delivery Route Optimisation

RKMVERI

Oct 2025 – Dec 2025

Implemented and evaluated deterministic and stochastic hill climbing algorithms for a drone delivery route optimization problem, modelling travel cost using both Euclidean distance and node-level congestion across 120 delivery locations. Deterministic hill climbing achieved the best route (112.36 hours), while stochastic hill climbing improved search exploration at the cost of longer routes (138.90 hours). Formulated the problem as a combinatorial optimization over permutations of delivery nodes, defining edge congestion as the mean of adjacent node congestion factors. Analyzed convergence behavior across algorithms, finding that deterministic hill climbing converged faster but was more sensitive to initial route configuration, while stochastic hill climbing traded convergence speed for broader exploration of the search space.